

EchoFrame and Ghostnet: Behavioral Spoof Simulations of Entropy-Based Identity Systems

Abstract

This whitepaper presents a fully sandboxed simulation of a synthetic Secure Element-based payment protocol dubbed 'PlumPay', paired with two proprietary behavioral cryptographic frameworks: Ghostnet and EchoFrame. Through AppleScript and localized shell-based entropy injections, the simulation recreates predictive spoof payloads using quantum-noise-floor inspired randomness, UID echo drift modeling, and synthetic cryptographic streams. EchoFrame predicts and constructs forward-drifting payloads that mimic authentic signatures prior to their real issuance.

Critically, the simulation does not perform any live transmission or injection into NFC fields or BLE channels. All behavior is rendered through internal loop emulation and log generation.

This project demonstrates a potential vulnerability class where high-fidelity simulations might be weaponized if left unregulated. While PlumPay remains fictional, the architectural proof and behavioral modeling could inform next-generation payment system defense, especially against entropy spoofing and identity drift exploits.

This document is for cybersecurity research purposes only and includes zero operational payloads, exploits, or unauthorized transmissions.